



BCT UNIT 3 - Summary computer science

computer science (Vignan's Institute of Engineering for Women)



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UNIT-3

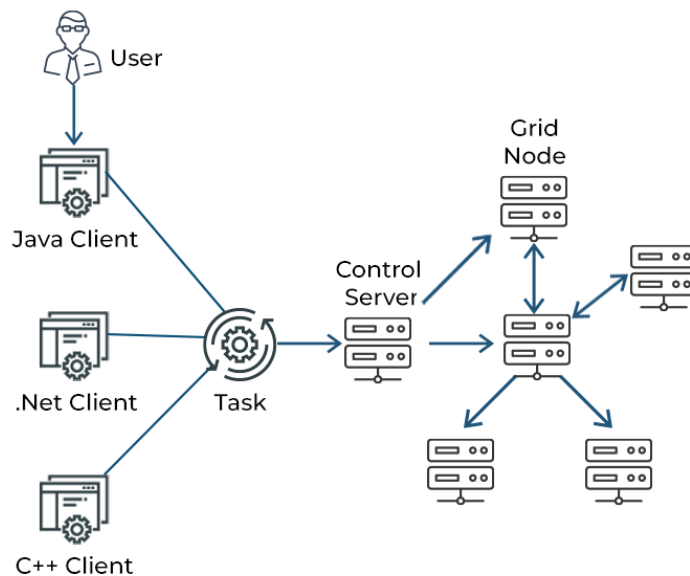
Blockchain technology is most simply defined as a decentralized, distributed ledger that records the provenance of a digital asset.

Grid coin

Gridcoin is an open-source blockchain that mints and distributes cryptocurrency. It is secured through a proof-of-stake protocol and monitors processing contributions to BOINC. Gridcoin can be bought and sold in exchange for conventional currencies, such as the USD, or it can be traded on cryptocurrency exchanges.



HOW GRID COMPUTING WORKS



How do I get Gridcoin?

To earn Gridcoin, you need to crunch whitelisted BOINC projects on teamGridcoin. You must install the BOINC client and select several BOINC projects. Forgetting Gridcoin, see step 3. You can also start out by pool mining, which needs no Gridcoin to get rewards.

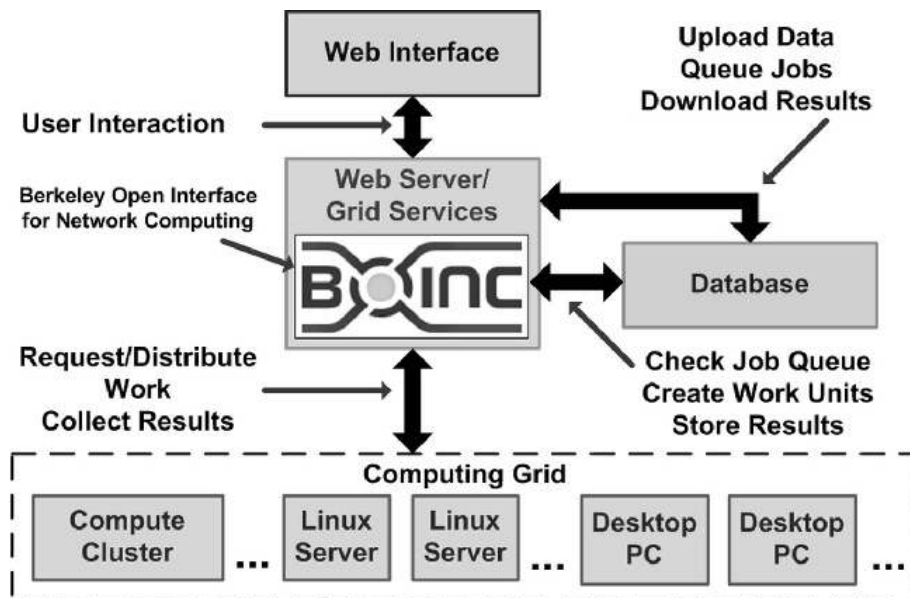
Worth of Grid coin:

• GridCoin price today is \$0.009239 USD with a 24-hour trading volume of \$329.64 USD. About GridCoin. GridCoin Price \$0.009239 USD.

How and where to buy Grid Coin(GRC)

- Step 1:BuyBTC or ETH at Coinbase. Sign up andpurchaseBTC or ETH at Coinbase.
- Step 2: Go to a supporting GRC exchange: Transfer your newly purchased BTC or ETH from your Coinbase wallet to one of the exchanges listed below. OpenLedger DEX.BUY. C-Cex.BUY. CoinSwitch

.BUY. ALL GRC EXCHANGES.Is crypto is a Currency?



•Cryptocurrency is digital money. That means there's no physical coin or bill — it's all online. You can transfer cryptocurrency to someone online without a go-between, like a bank. Bitcoin and Ether are well-known cryptocurrencies, but new cryptocurrencies continue to be created.

Advantages

- Can solve larger, more complex problems in a shorter time
- Easier to collaborate with other organizations
- Make better use of existing hardware

Disadvantages □

Grid software and standards are still evolving □ Learning curve to get started □ Non-interactive job submission

Folding Coin:

FoldingCoin (FLDC) is a digital token that compensates participants (folders) for their Folding@home (FAH) computational power. With growing community support, more folders are consistently joining the FAH network, to help find cures for cancer, Alzheimer's, and many other viral diseases. With the Merged Folding platform, you can earn more than just FLDC, but also other cryptocurrencies.

To start earning FLDC, please use our FOLDING BROWSER if you are on Windows, as this will ensure that you are setup correctly to receive FLDC and our partner coinCURECOIN



FoldingCoin

Mine Medicine, Not Hashes

WHAT IS FOLDING@HOME?

Folding@home(FAH) is a project by Stanford University that has been running since October 2000. FAH uses idle computer power to help simulate how proteins fold in the human body. This research is then used to help researchers find cures for diseases such as cancer and Alzheimer's.



Researchers normally have to spend millions of dollars on Supercomputers to do these simulations. With FAH and the power of the internet, the workload is broken up into

small Work Units (WU) and sent to anyone who downloads and runs the FAH program on their computer. This allows Stanford to get the research done without the need of Supercomputers, but rather with the help of many individuals running the program. What are the benefits of FoldingCoin?

•FoldingCoin Technology

- FoldingCoin runs the PHP code on its commercial hosting provider to keep track of the FLDC coins awarded to the users. The code awards 2500,000 FLDC coins in a day in proportion to the FAH credits earned by a user in a period of 24 hours.

- FLDC also uses **LetsTalkBitcoin!** (LTB) to build a crypto space for its user community. LTB has authorized the FoldingCoin platform to use its automated distributor facility to improve upon its transactional reward processes.

- The project also uses **LTB's Token Controlled Access** (TCA) platform to provide a communication platform to its participants. TCA creates private forums that allow the users to hold conversations that are accessible to only those who have FLDC funds

WHAT DOES FOLDINGCOIN DO?

FoldingCoin provides an incentive to Folders by distributing FLDC and other tokens, based on the amount of computational power contributed to FAH. Stanford has a **STATS PAGE** that tracks all the work being completed by each individual folder. Folders earn "points" by completing WUs in a timely manner. At the end of each month, the FoldingCoin organization downloads the stats of each folder and distributes FLDC proportionate to the amount of completed work.

FoldingCoin's primary objective is to contribute to the humanitarian cause of the FAH project by substantially increasing the FAH network by providing an incentive-based rewards. FLDC is built on the **Counterparty** protocol which allows for tokens to be built within the Bitcoin blockchain. So every FLDC transaction is actually a Bitcoin transaction as well. FLDC is shared in the same wallet that you can hold Bitcoin and other Counterparty tokens in. Earning FLDC can help offset the electricity cost and bring motivation to Folders to buy bigger and better Folding Rigs.

Features of Folding coin?

- Common hardware
- Counter party
- Ease to setup
- Large user base

- Nonprofit

How to Buy and store FLDC?

•FoldingCoin's crypto currency FLDC is available for purchase on exchanges likePoloniex, Bittrex, Counterparty, UpBitandHuobi.

• CoinSwitch, asthe largestcryptocurrencyconversion platform, offers you the ability to exchange your cryptos for FLDC. You have the option to go through over 45,000 pairs of crypto exchange to choose your preferred one.

•What's more - with CoinSwitch, you do not rely on a separate wallet from the exchange to store your FoldingCoin cryptocurrency. Instead, all transactions happen to and from your own personal private wallets.

Storage:

•FoldingCoin's own wallet is a reliable means of storage for FLDC coins. Users can download the wallet from its official website.

•Furthermore, FLDC in an ERC20 compliant coin. One of the most suitable options for storing FLDC coins, therefore, is the myEtherWallet. You have to first visit the website and create an account, if you don't already have one. Download the keystore file and back it up at a safe location. You will then receive a private key to your wallet. Get a printout of this key for safety purposes.

Blockchain Genomics:

[Blockchain technology](#) has enormous potential to build the systems of the future and shape the medical world in new ways. A blockchain is a digital, decentralized public ledger designed to record every data on its network. Blockchain is widely deployed in various sectors, such as finance, education, healthcare, etc.

- Blockchain technology may be new, but it is already playing a major role in healthcare, especially in genomic medicine.
- As the technology becomes more widely used in genomics, Scientists will be able to gain a deeper understanding of disease mechanisms.
- This knowledge will help scientists develop better therapies and interventions for a variety of diseases while maintaining privacy and security.

Why Blockchain in Genomics?

Below are some of the points to understand why blockchain is useful in genomics:

- **Genomic Data Security:** Genomics data is very sensitive and crucial, from a data security perspective, blockchain provides excellent data security and integrity. Security methods such as encryption are useful in combating data breaches, but they do not provide complete protection. Many systems of big organizations with the highest level of security are penetrated by hackers. However, blockchain technology helps organizations by providing better protection against data breaches. Blockchain uses hashing techniques to store data securely, which helps the company in securing data and also helps in data sharing.
- **Genomic Data Sharing:** Using genomic data and the blockchain network, it is now possible to send anonymous genetic information around the world. The decentralization nature of blockchain allows easy and secure data exchange between organizations. Information can be stored in a special ledger in a blockchain database that keeps the information secure.
- **Immutability of genomic data:** Blockchain provides immutability of genomic data for organizations, helping organizations protect information. Due to the decentralized structure of blockchain technology, genomic data cannot be modified, so any changes will be reflected on all nodes, so no one cheats here, and it can be said that genomic data sharing is very safe.
- **Efficiency:** The organization uses blockchain technology for efficiency reasons, as blockchain eliminates any third-party interference between genomic data sharing and errors, making the system more efficient and faster. As a result, sharing data becomes easier, smoother, and faster.
- **Cost Reduction:** Since blockchain does not require a third person, it reduces costs for organizations and gives trust to other partners. Before blockchain technology, organizations spend lots of money, as they have to hire a third person to maintain all the things which blockchain technology does.

Why Blockchain Technology Suitable For Bioinformatics and Healthcare Applications?

Blockchain plays an important role in bioinformatics and health applications as it reduces the cost of analysis in genomics and health applications.

- Transactions of data on blockchain systems are faster and more efficient than traditional processes.
- The biggest advantage of the blockchain is that the data recorded in the central system can be manipulated, while the data in the blockchain cannot be manipulated. Since it is a distributed system, every data transaction with the blockchain becomes more transparent and traceable.
- Blockchain prevents fraud and unauthorized activity better than other systems.

Why Blockchain Technology Not Suitable For Bioinformatics and Healthcare Applications?

- In genomics and healthcare, the biggest problem is that people don't want to share the raw data directly because, with the help of entire genome data, people can be detected and information about their ancestry can be obtained.
- In blockchain transactions, data is stored in its hash value rather than the original data.
- Neither current nor blockchain-based genomics and healthcare systems can provide complete anonymity until people use wearable devices to obtain their own analysis results.

Current Trends in Blockchain-Based Systems in Genomics

Let's take a look at the era of blockchain-based systems to better understand current trends. According to the NLM (National Library of Medicine) paper, the era of blockchain-based systems in genomics is divided into three eras, as follows:

1. Proof of Concept Era (From 2016 To 2018): Since the blockchain concept is new to the field in this era, the proposed application requires core development. The main goal of this era is to provide an efficient application to prove that the blockchain platform can be used for health management systems. In this era, blockchain platforms have:

- Include genomic data sharing and EHR sharing, while in other era research are conducted only for EHR sharing.

- In this era, the detailed architecture of the blockchain-based healthcare system was designed, while in other eras, the proposed method integrated other technologies into the system by reducing the blockchain part over time.

2. Blockchain Development Era (From 2019 To 2020): In this era, complex cloud-based and encryption-based applications are mainly proposed. The main research areas during this period were:

- The research focuses on designing a blockchain-based healthcare system, as in the first era.
- The research focus is not on disease prediction and patient monitoring methods, but on the use of various cryptographic techniques in the system to increase the security of the system, the implementation of blockchain-enabled modules, and evaluation of performance.

3. Blockchain As a Platform Era (From 2021 To Present Day): In this era, blockchain is converted into a platform that has additional AI-based algorithms running on the blockchain platform. This period can be considered the first period in building a data ecosystem using blockchain technology. The main research areas during this period were:

- The research focus is on the development of a blockchain-based healthcare system with patient monitoring and disease prediction capabilities.
- Artificial intelligence methods have been integrated into the system to pay more attention to these issues and evaluate the performance of the system from this perspective

Bit Coin MOOCs

The acronym MOOCs stands for Massive Open Online Courses. These are online courses that are designed to be accessible to a large number of learners from around the world. MOOCs are typically open to anyone who wants to enroll, and they often provide free or low-cost access to educational content.

When you refer to "Bitcoin MOOCs," you are likely talking about Massive Open Online Courses that focus on Bitcoin and related topics. These courses are designed to educate individuals about Bitcoin, its technology, its applications, and the broader

blockchain and cryptocurrency ecosystem. Bitcoin MOOCs can cover a wide range of topics, including:

Introduction to Bitcoin: An overview of what Bitcoin is, its history, and its basic principle

Bitcoin Technology: Explaining the underlying technology of Bitcoin, including blockchain, mining, and cryptographic concepts.

Bitcoin Economics: Discussing the economic aspects of Bitcoin, such as its monetary policy, supply cap, and potential impact on the financial system.

Bitcoin Trading and Investment: Providing information on how to buy, sell, and invest in Bitcoin and other cryptocurrencies.

Bitcoin Security: Teaching users how to safely store and manage their Bitcoin holdings to protect against theft and hacking.

Bitcoin and Legal/Regulatory Issues: Addressing the legal and regulatory aspects of Bitcoin in different countries and regions.

Bitcoin Development: Covering topics related to Bitcoin software development and the open-source nature of the Bitcoin network.

Bitcoin MOOCs are valuable resources for individuals who want to gain a deeper understanding of Bitcoin and cryptocurrency, whether for personal interest, academic study, or professional development. They are often offered by universities, online learning platforms, or specialized cryptocurrency organizations.